Lung Cancer Research Summary — For Family Reference

Prepared: April 3, 2026
For: Michael's mother-in-law (late stage lung cancer)

⚠️ **Important:** This is informational only, NOT medical advice. All treatment decisions should be made with her oncology team. Palliative care consultation is strongly recommended for symptom management.

🎯 Current Treatment Advances (2025-2026)

1. Antibody-Drug Conjugates (ADCs)

- **What:** Targeted therapy combining monoclonal antibodies with cytotoxic drugs
- **Status:** FDA approved 2025 for NSCLC (non-small cell lung cancer)
- **Benefit:** More precise targeting, fewer side effects than traditional chemo
- **Ask her oncologist about:** trastuzumab deruxtecan, sacituzumab govitecan

2. Immunotherapy Advances

- **PD-1/PD-L1 inhibitors:** Pembrolizumab, nivolumab, atezolizumab
- **New combinations:** Immunotherapy + targeted therapy showing improved survival
- **BioNTech pipeline:** Presenting new data at ELCC 2026 (March 2026)
- **Best for:** Patients with specific biomarkers (PD-L1 expression)

3. Targeted Therapies

- **For specific mutations:** EGFR, ALK, ROS1, KRAS G12C
- **2025 FDA approvals:** Two new targeted therapies for NSCLC
- **Testing needed:** Comprehensive genomic profiling of tumor

4. Small Cell Lung Cancer (SCLC)

- **New strategies:** Immunotherapy combinations showing promise
- **Barrier:** Still limited options compared to NSCLC
- **Clinical trials:** May offer access to emerging treatments

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🌿 Complementary Approaches (Evidence-Based)

Nutrition & Diet

What Helps:

- **Plant-based emphasis:** Higher vegetable/fruit intake associated with better treatment response (Springer Nature 2025)
- **High protein:** 1.2-1.5g/kg body weight to prevent muscle wasting
- **Omega-3 fatty acids:** Fish oil may reduce inflammation, improve appetite

- **Small frequent meals:** Better tolerance than 3 large meals

What to Avoid:

- **Ultra-processed foods:** Associated with increased lung cancer risk (Thorax BMJ 2025)
- **Excessive sugar:** May promote inflammation
- **Alcohol:** Can interfere with treatment, dehydrate

Specific Foods for Symptom Relief:

- **Ginger tea:** Nausea relief
- **Honey + warm water:** Soothes throat, helps with phlegm
- **Bone broth:** Easy protein, hydration
- **Avocado, nut butters:** Calorie-dense if appetite is poor

Mind-Body Practices

- **Acupuncture:** Evidence for nausea, pain, fatigue management
- **Gentle movement:** Walking, tai chi (if energy permits)
- **Breathing exercises:** For cough/phlegm management

💊 Symptom-Specific Management

1. Phlegm/Mucus (Most Common)

Medical:

- **Mucolytics:** Guaifenesin, N-acetylcysteine (NAC)
- **Nebulized saline:** Helps loosen secretions
- **Prescription options:** Atropine drops, scopolamine patch (for excessive secretions)

Non-Pharmacological (Hong Kong study 2026):

- **Humidifier:** Keep air moist, especially at night
- **Steam inhalation:** Bowl of hot water, towel over head, 5-10 min
- **Hydration:** Warm fluids throughout day
- **Chest physiotherapy:** Gentle percussion on back (ask palliative care to demonstrate)
- **Elevated sleeping:** Prop up with pillows
- **Honey + lemon:** Soothes, helps clear throat

Avoid: Dairy if it seems to thicken mucus (individual response varies)

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2. Right Leg Swelling (Edema)

Possible Causes:

- **DVT (Deep Vein Thrombosis):** ⚠️ **URGENT** — cancer increases clot risk 7x
- **Lymphatic obstruction:** Tumor pressing on lymph vessels
- **Low albumin:** From poor nutrition
- **Medication side effect:** Some drugs cause edema

Red Flags (Seek immediate care):

- Sudden swelling
- Pain, warmth, redness in leg
- Shortness of breath (could indicate pulmonary embolism)
- One leg much larger than other

Management:

- **Compression stockings:** 20-30 mmHg (if DVT ruled out)
- **Elevation:** Legs above heart level when resting
- **Gentle movement:** Ankle circles, short walks
- **Reduce salt:** Helps with fluid retention
- **Diuretics:** Only if prescribed (can worsen some situations)

Ask oncologist: "Could this be DVT? Should we do an ultrasound?"

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3. Shoulder Itch (Pruritus)

Possible Causes:

- **Paraneoplastic syndrome:** Cancer-related itch (documented in lung cancer)
- **Nerve involvement:** Tumor pressing on nerves (Pancoast tumor)
- **Dry skin:** From treatment, dehydration
- **Medication side effect:** Opioids, some chemo drugs

Relief Strategies:

Topical:

- **Moisturize frequently:** Fragrance-free cream (Cetaphil, CeraVe)
- **Cooling gels:** Menthol or camphor-based
- **Calamine lotion:** Soothing
- **Hydrocortisone 1%:** For localized itch (short-term)

Systemic:

- **Antihistamines:** Cetirizine (non-drowsy), diphenhydramine (at night)
- **Gabapentin:** For nerve-related itch (prescription)
- **Mirtazapine:** Low dose helps itch + appetite + sleep (prescription)

Non-Pharmacological:

- **Cool compresses:** 10-15 min on itchy area
- **Oatmeal baths:** Colloidal oatmeal in lukewarm water
- **Loose cotton clothing:** Avoid wool, synthetics
- **Keep nails short:** Prevent skin damage from scratching
- **Distraction:** Gentle massage elsewhere, music, conversation

Ask palliative care: "Could this be paraneoplastic pruritus? What's the best approach?"

🏥 Palliative Care Recommendations

Why Palliative Care (NOT just hospice):

- Can start at diagnosis, alongside treatment
- Focuses on symptom relief + quality of life
- Reduces hospitalizations, may extend survival
- Supports family emotionally

What to Request:

1. **Comprehensive symptom assessment**
2. **Medication optimization** (avoid polypharmacy)
3. **Home care support** if available
4. **Family education** on what to expect
5. **Emotional/spiritual support** for patient + family

In Singapore:

- HCA Hospice
- Assisi Hospice
- Palliative Care Centre @ NHG

⚠️ Approaches to Be Cautious About

Not Recommended (No Evidence or Harmful):

- High-dose vitamin C IV (can interfere with chemo/radiation)
- Laetrile/amygdalin (toxic, no benefit)
- Cannabis oil as "cure" (may help symptoms, not curative)
- Extreme diets (Gerson, alkaline) — can cause malnutrition
- Supplements without oncologist approval (many interact with treatment)

May Help Symptoms (Discuss with Doctor First):

- Medical cannabis: Appetite, nausea, pain, sleep
- CBD oil: Anxiety, sleep (evidence mixed)
- Turmeric/curcumin: Anti-inflammatory (absorption issues)
- Medicinal mushrooms: Immune support (quality varies)

📋 Questions for Her Oncology Team

1. "What type of lung cancer is it? (NSCLC vs SCLC, specific subtype)"
2. "Has comprehensive biomarker testing been done? (EGFR, ALK, ROS1, KRAS, PD-L1)"
3. "Are there targeted therapy or immunotherapy options we haven't tried?"
4. "Could the leg swelling be DVT? Should we ultrasound?"
5. "What's the best approach for the phlegm/cough?"
6. "Can we get a palliative care referral for symptom management?"
7. "Are there clinical trials appropriate for her situation?"

8. "What should we expect in coming weeks/months?"

💝 For the Family

What You're Doing Matters:

- Being present is the greatest gift
- Listening without fixing
- Creating moments of joy, however small
- Handling practical tasks so she can rest

Watch For (End-of-Life Signs):

- Decreased appetite/fluid intake
- More sleeping, less responsiveness
- Changes in breathing pattern
- Cool extremities, mottling
- Confusion or withdrawal

When These Appear:

- Focus on comfort, not nutrition/hydration
- Mouth care (swabs, lip balm) more important than food
- Speak softly, assume she can hear you
- Hold hands, play favorite music
- Call palliative care/hospice for guidance

📚 Resources

For Medical Information:

- American Cancer Society: cancer.org
- Lung Cancer Alliance: lungcanceralliance.org
- National Cancer Institute: cancer.gov

For Family Support:

- "Being Mortal" by Atul Gawande
- "The Five Invitations" by Frank Ostaseski
- Local hospice family programs

Compiled with care for Michael's family. May this information bring clarity and peace. 🪷

Last updated: April 3, 2026